Time Has a Price

Would you pay to keep waiting?

A mathematical decision lab using real market prices



A finite budget for decisions

100 cash + a limited time budget

Each market day uses 5 budget seconds.
A pack adds 15 seconds; the main price is 8 cash.

Exit
Continue with 0%, 50% or 100% invested
Buy a pack, then choose the same three exposures

$$U(B, L) = (1 - \alpha)\ln(1 + B/100) + \alpha \ln(1 + L/90)$$

B: cash after settlement · L: remaining budget · α: time preference
Budget seconds are model units, not measured human lifespan.

Real prices, explicit time costs

BTC 2022–2025

Frozen market model

BTC + ETH
January–September 2026

61,344 replay episodes

BTC baseline

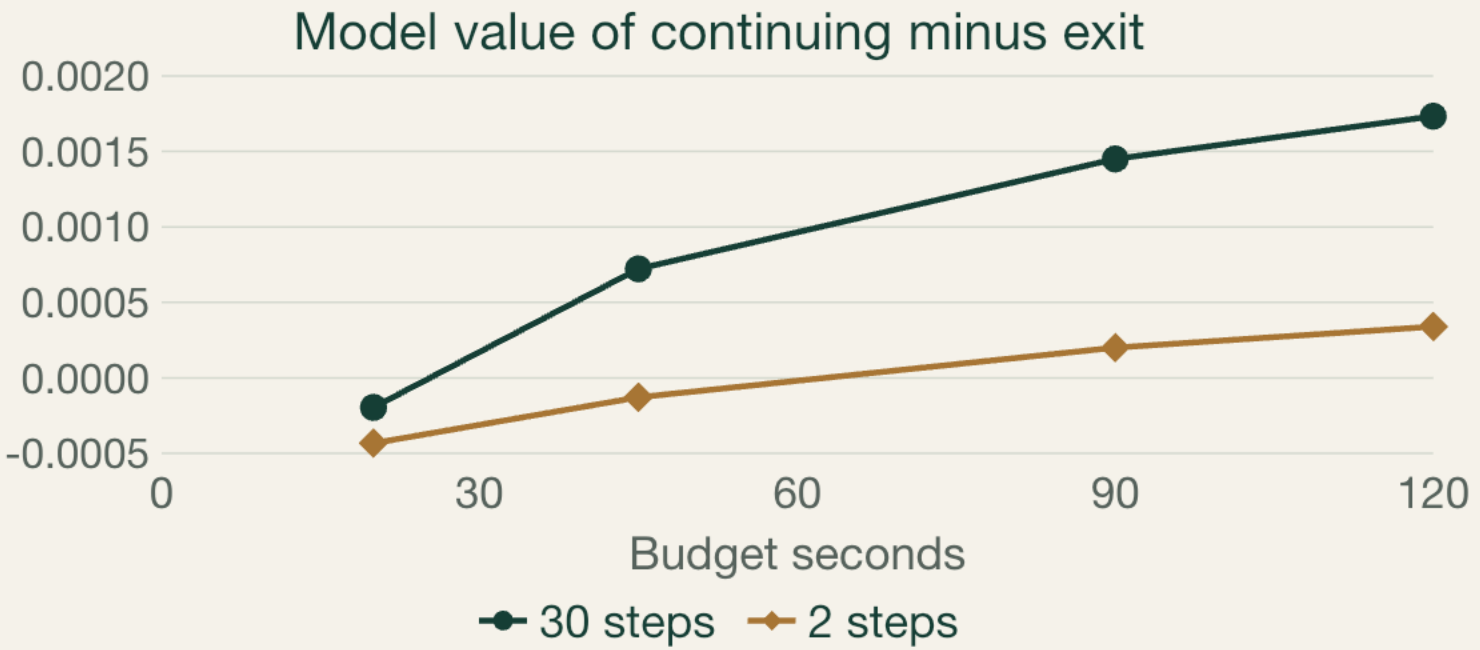
Strategy	Final cash	Budget left
30-step planner	100.00	90.00
2-step planner	100.00	90.00
EMA with a budget	100.09	5.00
EMA buying packs	85.34	8.05

Pack price 8 · α = 0.1 · initial budget 90
All constrained methods share fees and settlement rules.

Means across 213 overlapping starts per asset. We had already inspected 2026 data; this is historical re-evaluation, not a blind test.

A longer horizon can change the action

Same 100 cash · falling / volatile market · pack price 8 · α = 0.02



At 45 budget seconds

2 steps: EXIT

Continue – exit: –0.000129

30 steps: INVEST 100%

Continue – exit: +0.000722

6 of 192 initial decisions change. Historical improvement remains unconfirmed.

BTC realized utility difference: –0.000708 · 95% block interval [–0.003254, +0.001074]
The interval includes zero. The chart shows model expectations, not known future returns.

We built a 30-step optimizer. At the main setting, it tells us to leave.

The 30-step planner buys no extra time during trading at prices 8 or 2.

Real prices; assumed time costs and preferences.
The next test needs genuinely unseen data.

Python + NumPy + React · AI-assisted build
12 tests; 391,772 account-ledger events independently checked